



- . **M**obile
- . **A**utonomous
- . **P**ath-Following
- . **S**ystem

A1, Q2



Meet the Team



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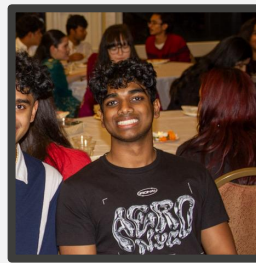
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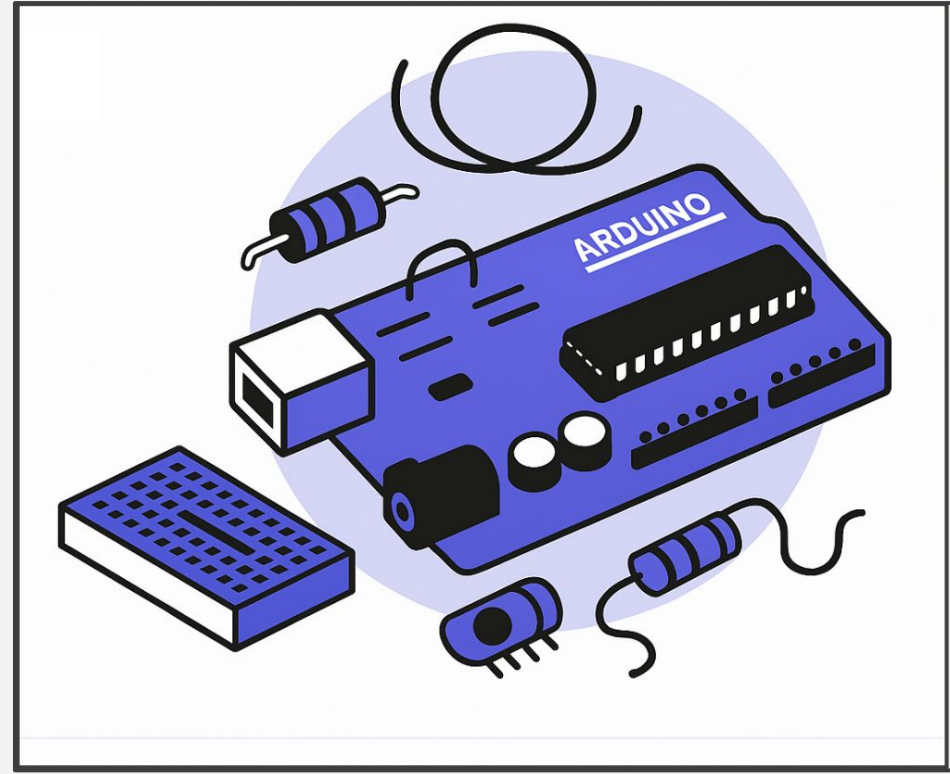
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Problem Statement

Design an autonomous, touchless-activated vehicle that follows a path, avoids obstacles, carries a load, and detects a finish line using integrated sensing and control systems.



Executive Overview

Core Function

Autonomous robot that follows black tape paths, avoids obstacles, and recognizes finish lines

Result

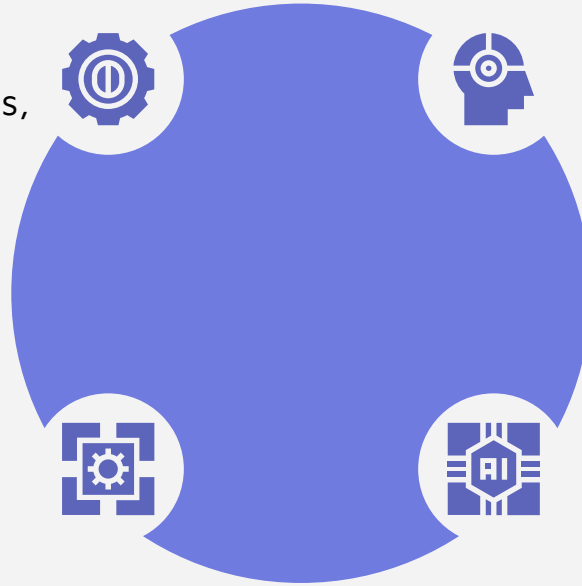
Successfully implemented embedded control and sensor fusion in working, modular prototype

Key Components

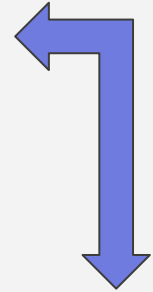
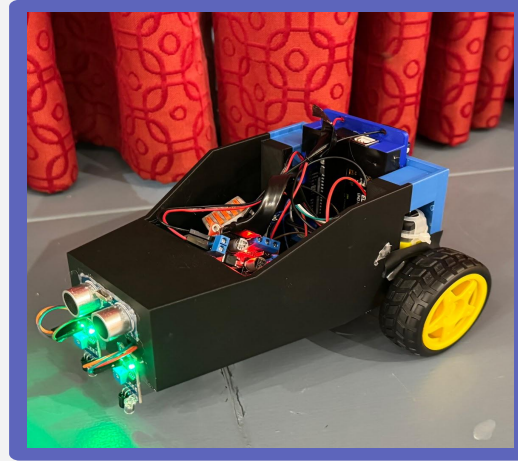
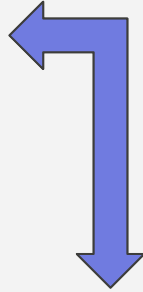
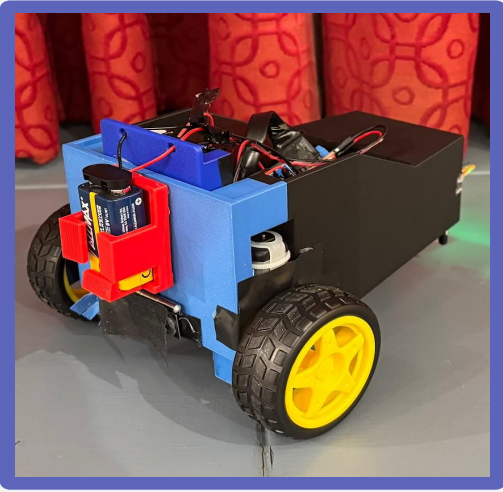
- Arduino Uno
- Dual IR sensors
- Ultrasonic sensor
- PWM motor control

Features

- Touchless activation system
- 250g payload capacity
- Obstacle detection, resume when obstacle gone
- Finish line stop



MAPS Design



Hardware Components:

- 3D-Printed PLA Chassis
- Rubber Wheels
- Battery Holder (for 9V)
- Electrical Tape

Electronic Components:

- Arduino Uno R3
- L298N Motor Driver
- TT DC Motors
- IR Line Sensors
- HC-SR04 Ultrasonic Sensor



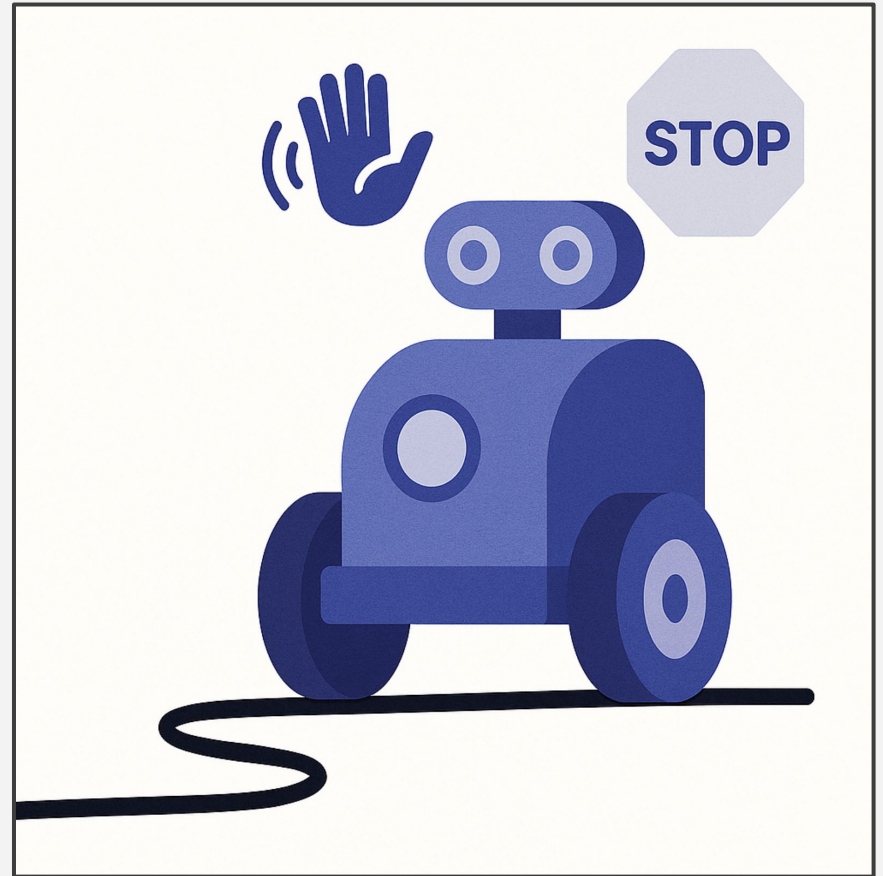
Usability Overview



User
Interfaces



Device
Operation



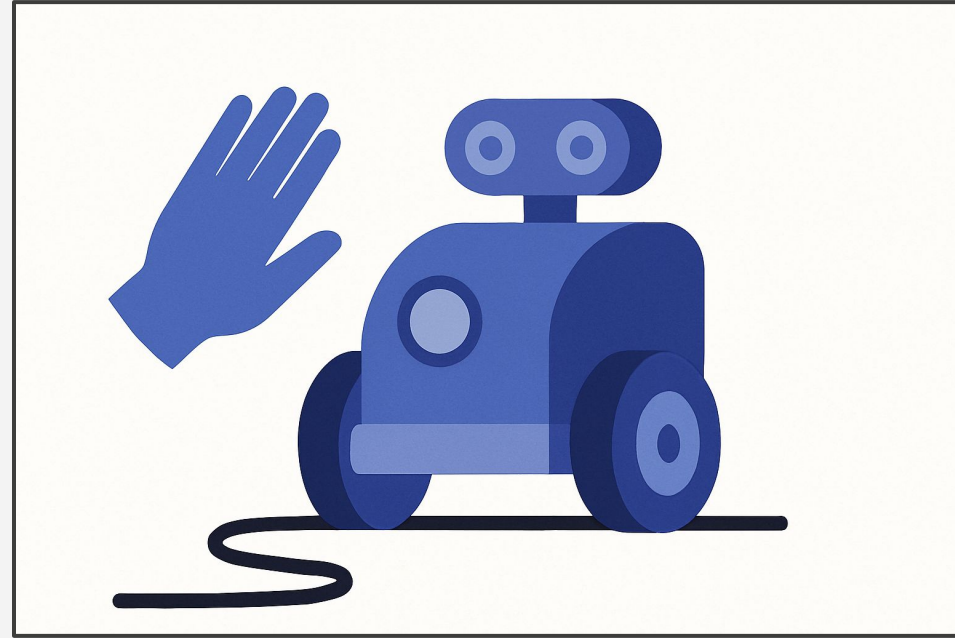


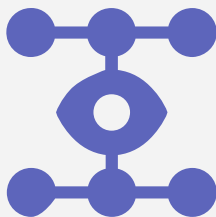
User Interface

**The UI for MAPS is designed
for ultimate simplicity**

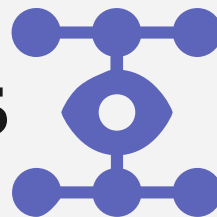


No buttons or switches involved. Once the battery is plugged in, the only action the user needs to take is a wave in front of the ultrasonic sensor





Device Operations



Sensor Logic

- 5ms sampling from 2 IR sensors
- Follows black tape on white background
- Finish line detected at 4cm width



Physical Conditions

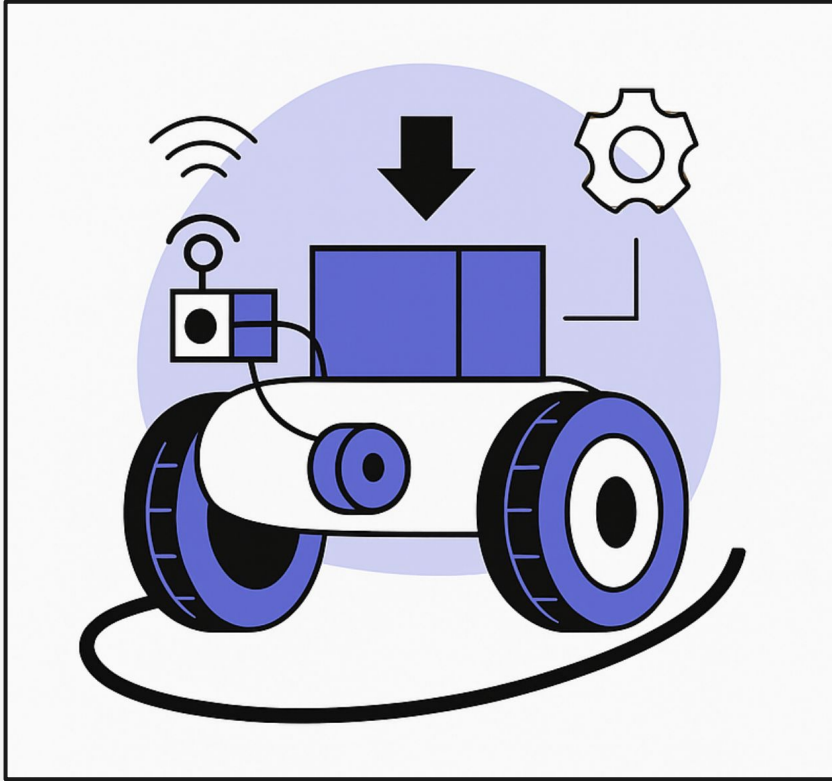
- Indoor lighting: 200–800 lx
- Flat, smooth surfaces
- Navigates up to 135° turns



Obstacle Handling

- Ultrasonic sensor triggers pause < 10 cm
- Auto-resumes once path is clear
- 95% success rate in trials



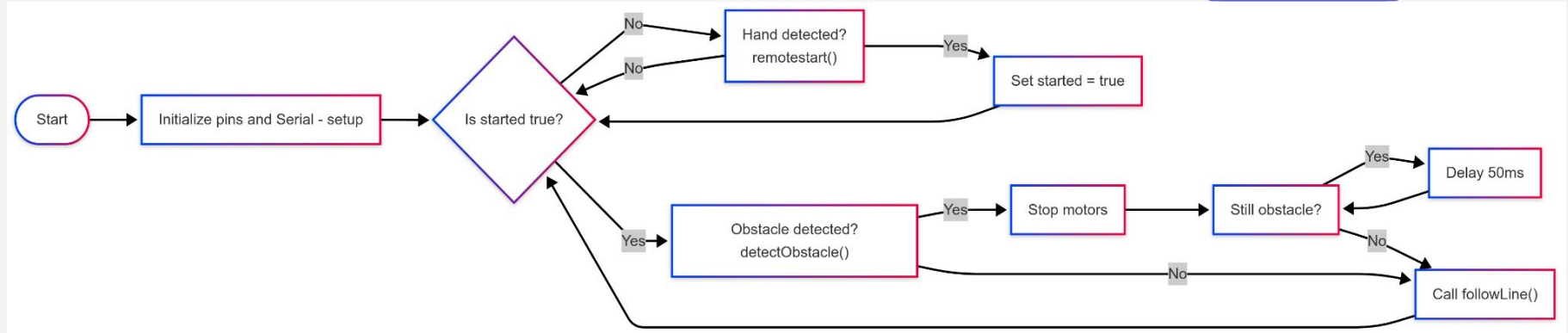
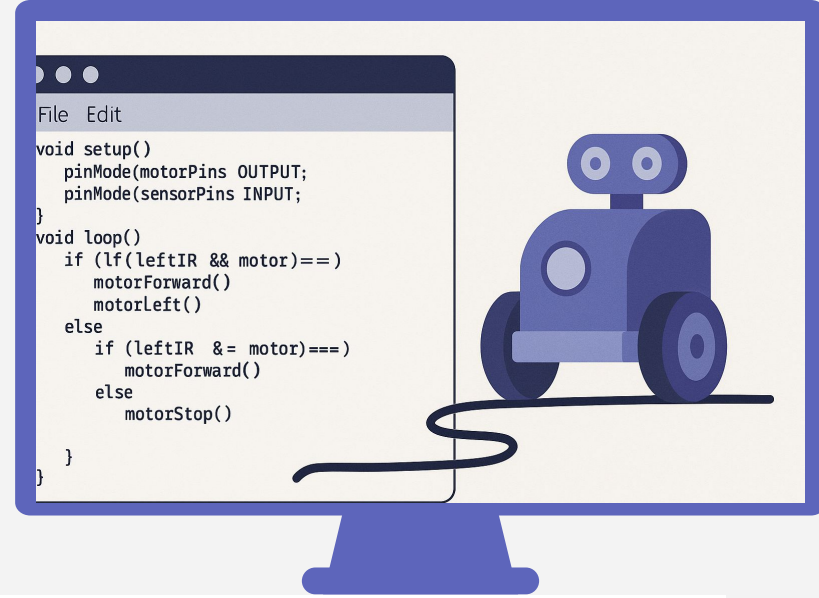


Code & Logic

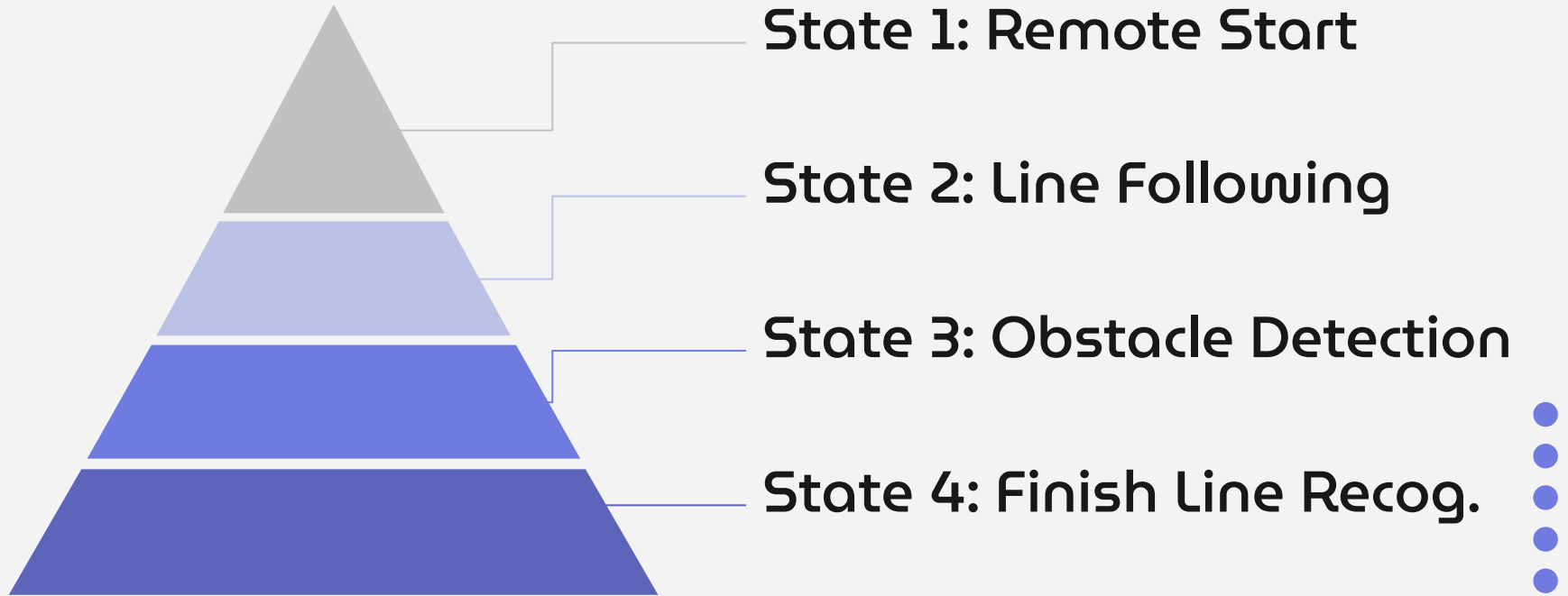


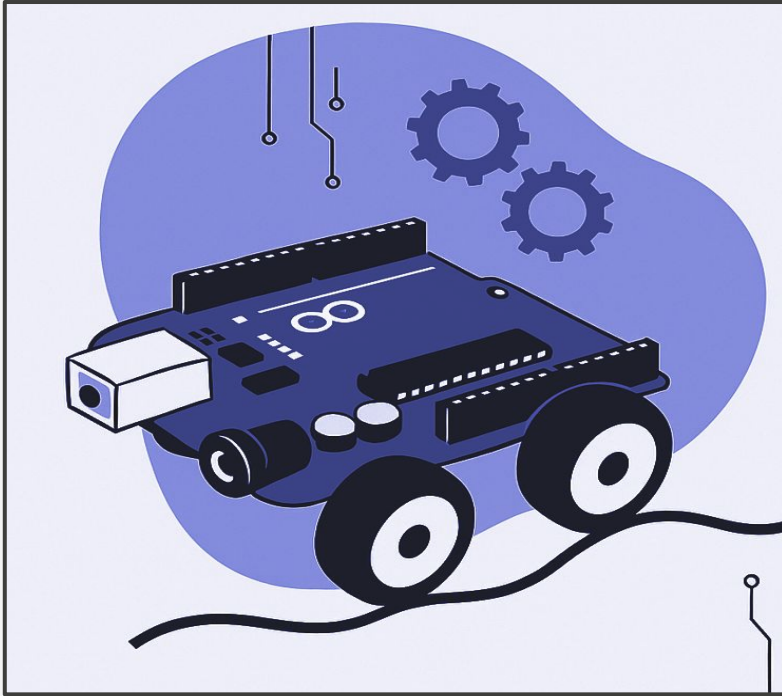
Simplified Code Flowchart Centered on the Logic

- More technical code flowchart can be found in the appendix(Figure Q1)



State Function





Objectives & Constraints



Accurate Autonomous Line Following



Objective

Reliably follow a 2 cm-wide black tape line on a white taped background



Constraint

Floor's high reflectivity and potential lighting variations

Estimated Metrics:

- Average Line Deviation: ~0.5-1 cm from tape center
- Line Detection Success Rate: >9/10 under normal indoor lighting

Precise Steering Control



Objective

Minimize zig-zag motion and smoothly navigate up to 135° turns



Constraint

Differential-drive with small DC gear motors (~7 cm wheel diameter)

Estimated Metrics:

→ Steering Overshoot: $<10^\circ$ in sharp turns

Obstacle Detection & Avoidance



Objective

Detect and stop before obstacles



Constraint

Limited sensor space and single
9V battery power budget

Estimated Metrics:

- Detection Range: 10 cm using Ultrasonic sensor
- Avoidance Success Rate: >9/10 in controlled indoor environments

Fast Response Time



Objective

Quickly adjust to abrupt line direction changes



Constraint

Single 9V battery (~500 mAh) and limited motor torque

Estimated Metrics:

- Control Loop Frequency: ~10–20 Hz (sensor-read → compute → motor-update).
- Reaction Time: <250 ms for sudden course corrections.

Finish Line Recognition



Objective

Detect the designated black tape finish line to stop and trigger an action



Constraint

Has to rely on the same line sensors

Estimated Metrics:

→ Detection Accuracy: 9/10

→ False Stop Rate: 1/20

Lightweight & Portable



Objective

Easy to transport



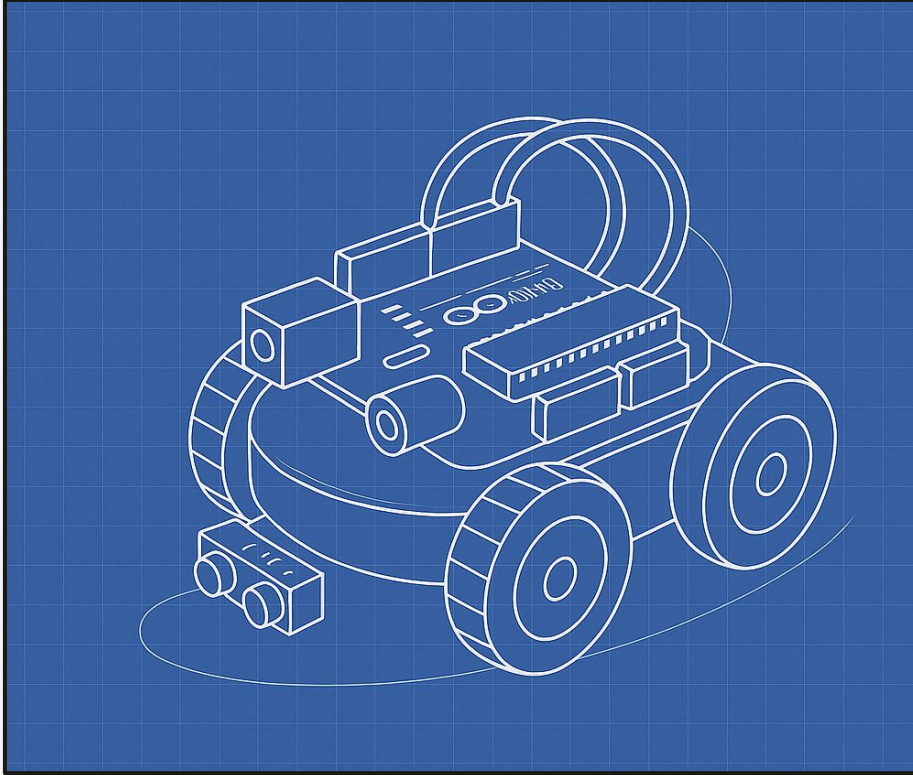
Constraint

Similar size as an 8.5x11 in. sheet of paper

Estimated Metrics:

→ Weight: ~0.5 lb total.

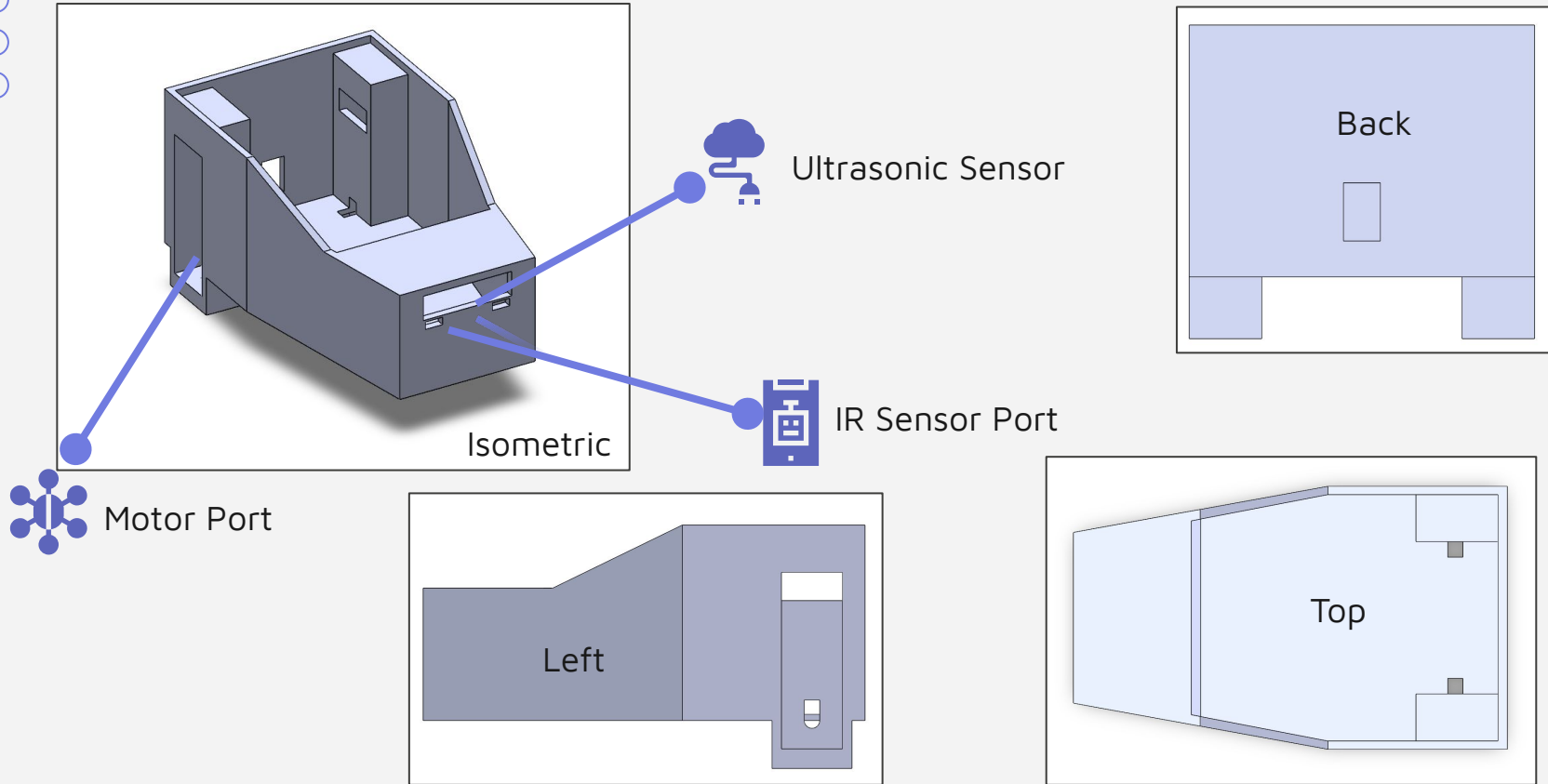
→ Single-Hand Carry ~ small form factor and minimal mass.

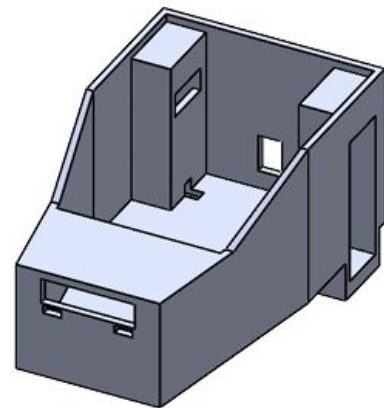
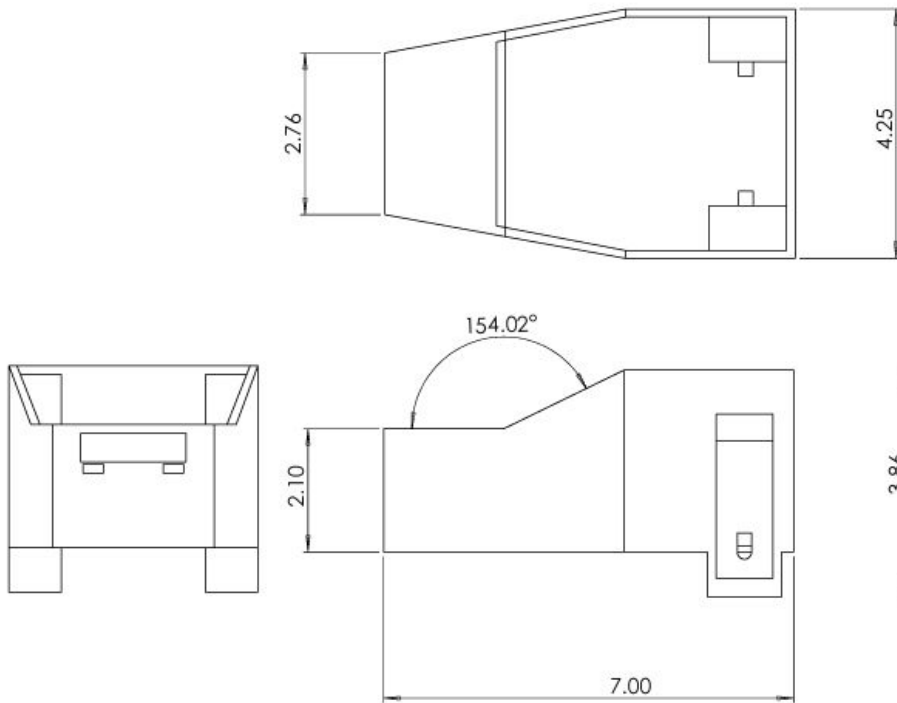


Drawings



Orthogonal Chassis Views





NOTES:
MEASUREMENTS ARE INCHES

A custom-built robot is shown from a high-angle perspective. The robot has a black, angular body. On its front, there are two blue sensor modules, each with a circular lens. Various colored wires (red, black, orange, green) are connected to the robot's electronics. A large black wheel with a yellow rim is visible on the left side. The robot is positioned on a speckled floor, and a white rectangular object is partially visible behind it.

Matthew, Andrew, Parth, Swas
A1 G2

Acknowledgement

Thank you to our professors,
friends, TAs, classmates, EPIC
employees, and Dall-E.

Questions?



Appendix

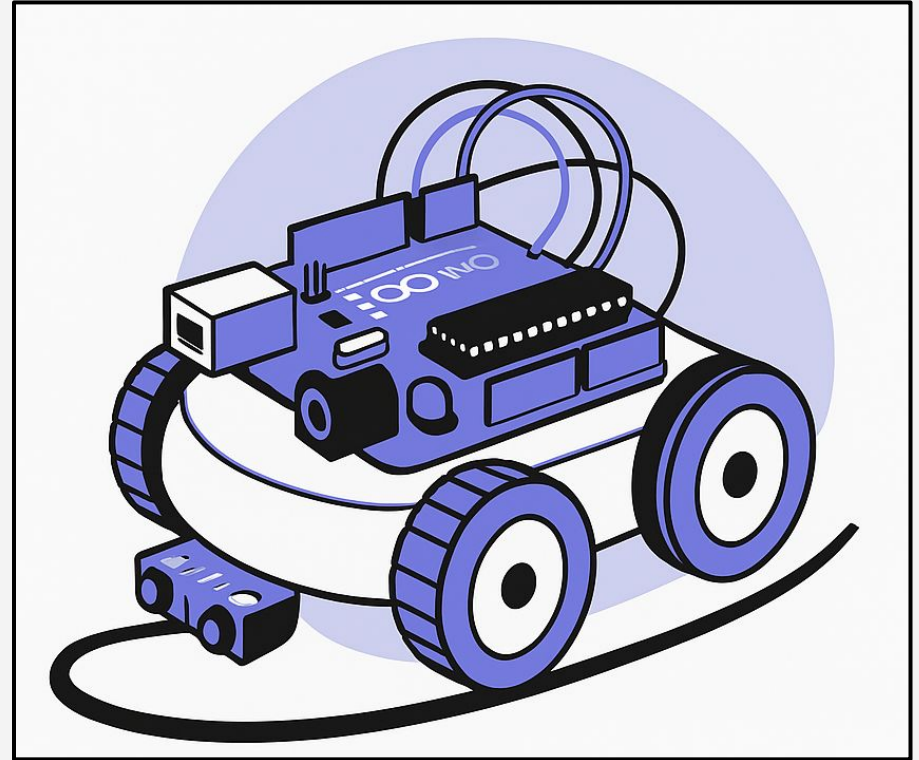
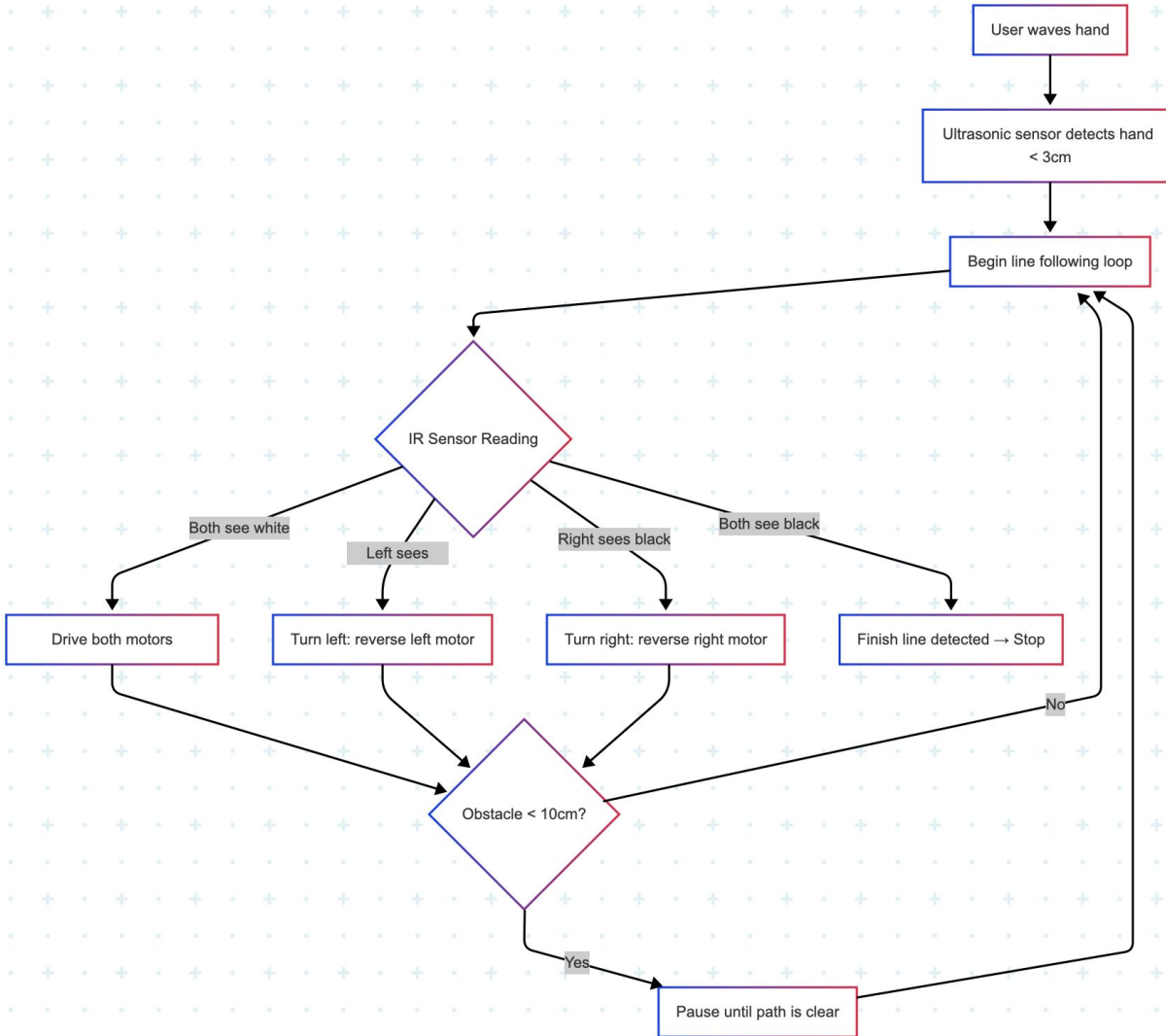




Figure A1: MAPS Operational Flowchart

Description:

This flowchart outlines the **core operational logic** of the MAPS. The process begins with a **touchless user interface**, initiated by a hand wave detected by the ultrasonic sensor. Once activated, the **robot enters a continuous line-following loop using dual IR sensors**. Based on real-time sensor input, the vehicle either drives straight, turns left, turns right, or stops at the finish line. Additionally, an **obstacle detection routine pauses the robot if an object is within 10 cm**, resuming motion once the path is clear. This logic enables hands-free, fully autonomous operation with responsive environmental handling.



	Accurate Autonomous Line Following	Obstacle Detection & Avoidance	Finish Line Recognition	Fast Response Time	Lightweight and Portable	Precise Steering Control	Score
Accurate Autonomous Line Following	x	1	1	1	1	1	5
Obstacle Detection & Avoidance	0	x	1	1	1	0	3
Finish Line Recognition	0	0	x	0	1	0	1
Fast Response Time	0	0	1	x	1	0	2
Lightweight and Portable	0	0	0	0	x	0	0
Precise Steering Control	0	1	1	1	1	x	4



Figure B1: Pairwise Comparison Matrix of Design Objectives

Description:

This matrix **compares the relative importance of each project objective** in a pairwise fashion to determine which features most significantly contribute to the MAPS robot's overall functionality. **"Accurate Autonomous Line Following" ranks highest with a score of 5**, highlighting it as the **most essential feature** for system performance, followed by "Precise Steering Control" with a score of 4. This prioritization guided key design and testing decisions during development.



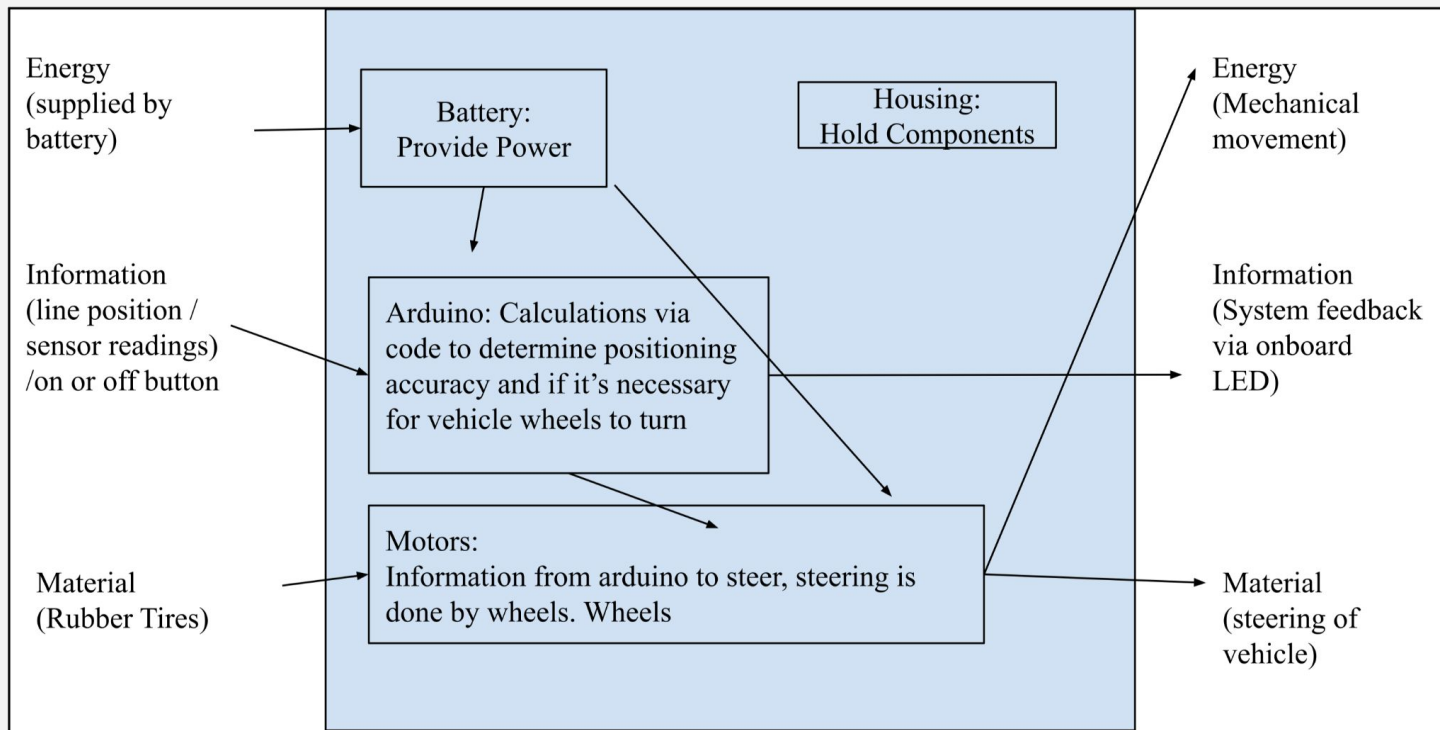


Figure C1: Glass Box Functional Diagram of MAPS

This diagram outlines **how energy, information, and material flow through MAPS**. The battery powers the Arduino, which processes sensor input and controls the motors for steering. The motors drive rubber tires to generate motion, while the housing holds all components. Outputs include movement, LED feedback, and steering control.



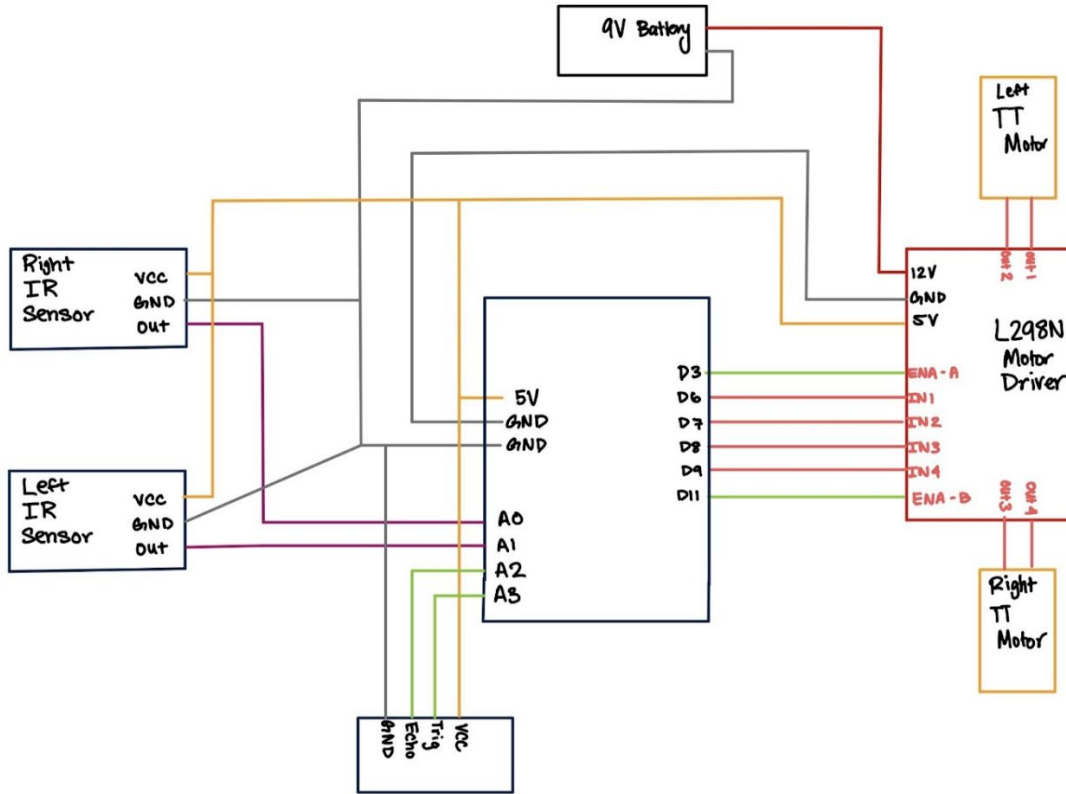


Figure D1: MAPS Circuit Diagram

Description:

This circuit diagram illustrates the **electrical connections between the key components** of the MAPS robot. The Arduino Uno serves as the central controller, receiving input from two IR sensors and an ultrasonic sensor, and outputting motor control signals to the L298N motor driver. The motor driver powers two TT DC motors and is supplied by a 9V battery. The **diagram shows how each sensor and motor is connected to specific digital and analog pins on the Arduino**, enabling real-time decision-making for autonomous navigation and obstacle avoidance.

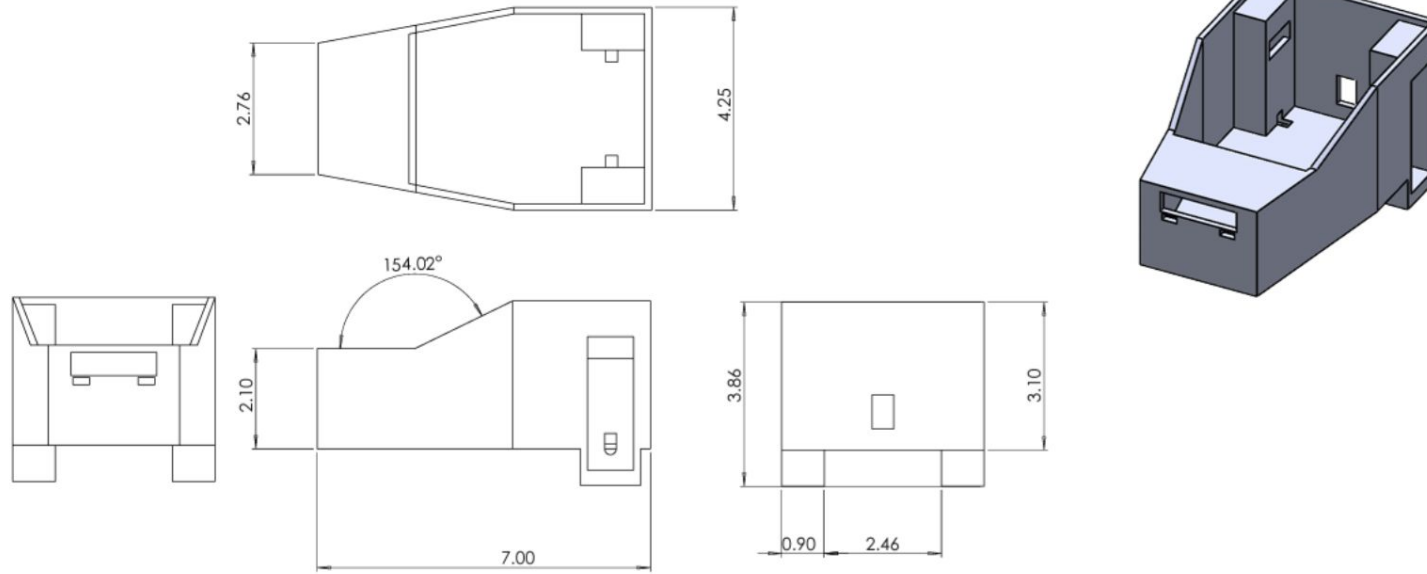


Figure E1: MAPS Chassis Technical Drawing

Description:

This technical drawing shows the **detailed dimensions and design of the 3D-printed MAPS chassis**. The layout includes top, front, side, and isometric views, highlighting component cutouts, slope angles, and mounting spaces. The housing measures 7.00 inches in length and features angled sidewalls for sensor visibility and internal slots to secure electronic components. This open-top design **improves accessibility during wiring, debugging, and component installation**.



Part	Quantity	Cost (\$)
KY-033 Line Following IR Sensor	2	3.04
Arduino Uno R3	1	8.99
9V Battery	1	1.60
HCSR04 USS Sensor	1	3.50
L298N Motor Driver	1	6.99
DC Gearbox Motor	2	5.90
Rubber Wheels	2	0.81
	Total	30.83



Figure F1: MAPS Bill of Materials (BOM)

Description:
This table lists the **components used in the construction of the MAPS robot along with their quantities and individual costs**. The total cost to build the device is **\$30.83**, making it a low-cost, educationally accessible platform. Key components include an Arduino Uno R3, KY-033 IR sensors for line detection, an HC-SR04 ultrasonic sensor for obstacle avoidance, a 9V battery for power, an L298N motor driver, DC gearbox motors, and rubber wheels.

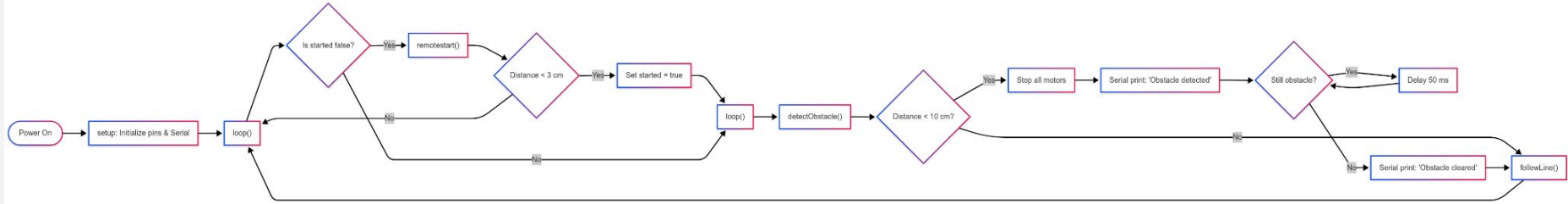


Figure G1: Technical Code Flowchart of MAPS Control Logic

Description:

This flowchart provides a detailed representation of the MAPS robot's control logic, as implemented in the Arduino code. It maps out the **initialization**, **remote activation using the ultrasonic sensor**, **obstacle detection routines**, and the **core line-following behavior**. Key function calls such as **remotestart()**, **detectObstacle()**, and **followLine()** are shown along with their decision points and system responses. The horizontal layout illustrates how the robot **transitions between idle, paused, and active navigation states in real-time**.

